

Movies Data Analysis

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Introduction

This report analyzes a dataset of movies, exploring genre popularity, budget versus box office performance, audience ratings, and trends over time.

1. Data Importing

```
movies_data <- read.csv("movies.csv")
head(movies_data)
```

```
##              name rating  genre year
## 1          The Shining      R   Drama 1980
## 2        The Blue Lagoon      R Adventure 1980
## 3 Star Wars: Episode V - The Empire Strikes Back    PG   Action 1980
## 4          Airplane!    PG   Comedy 1980
## 5        Caddyshack      R   Comedy 1980
## 6    Friday the 13th      R   Horror 1980
##      released score  votes      director
## 1 June 13, 1980 (United States)   8.4  927000    Stanley Kubrick
## 2  July 2, 1980 (United States)   5.8   65000     Randal Kleiser
## 3 June 20, 1980 (United States)   8.7 1200000     Irvin Kershner
## 4  July 2, 1980 (United States)   7.7  221000       Jim Abrahams
## 5 July 25, 1980 (United States)   7.3  108000       Harold Ramis
## 6  May 9, 1980 (United States)   6.4  123000 Sean S. Cunningham
##      writer      star      country budget  gross
## 1    Stephen King Jack Nicholson United Kingdom 1.9e+07 46998772
## 2 Henry De Vere Stacpoole Brooke Shields  United States 4.5e+06  58853106
## 3    Leigh Brackett   Mark Hamill  United States 1.8e+07 538375067
## 4     Jim Abrahams   Robert Hays  United States 3.5e+06  83453539
## 5   Brian Doyle-Murray   Chevy Chase  United States 6.0e+06  39846344
## 6    Victor Miller   Betsy Palmer  United States 5.5e+05  39754601
##      company runtime
## 1    Warner Bros.   146
## 2 Columbia Pictures   104
## 3      Lucasfilm    124
## 4 Paramount Pictures    88
## 5   Orion Pictures    98
## 6 Paramount Pictures    95
```

Explanation: read.csv() loads the dataset into a data frame called movies_data . head() shows the first 6 rows so we can confirm it loaded correctly.

2. Data Structure and Overview

```
str(movies_data)
```

```
## 'data.frame':    7668 obs. of  15 variables:
## $ name      : chr  "The Shining" "The Blue Lagoon" "Star Wars: Episode V - The Empire Strik
es Back" "Airplane!" ...
## $ rating    : chr  "R" "R" "PG" "PG" ...
## $ genre     : chr  "Drama" "Adventure" "Action" "Comedy" ...
## $ year      : int  1980 1980 1980 1980 1980 1980 1980 1980 1980 1980 ...
## $ released: chr  "June 13, 1980 (United States)" "July 2, 1980 (United States)" "June 20,
1980 (United States)" "July 2, 1980 (United States)" ...
## $ score     : num  8.4 5.8 8.7 7.7 7.3 6.4 7.9 8.2 6.8 7 ...
## $ votes     : num  927000 65000 1200000 221000 108000 123000 188000 330000 101000 10000 ...
## $ director: chr  "Stanley Kubrick" "Randal Kleiser" "Irvin Kershner" "Jim Abrahams" ...
## $ writer    : chr  "Stephen King" "Henry De Vere Stacpoole" "Leigh Brackett" "Jim Abrahams"
...
## $ star      : chr  "Jack Nicholson" "Brooke Shields" "Mark Hamill" "Robert Hays" ...
## $ country   : chr  "United Kingdom" "United States" "United States" "United States" ...
## $ budget    : num  1.9e+07 4.5e+06 1.8e+07 3.5e+06 6.0e+06 5.5e+05 2.7e+07 1.8e+07 5.4e+07
1.0e+07 ...
## $ gross     : num  4.70e+07 5.89e+07 5.38e+08 8.35e+07 3.98e+07 ...
## $ company   : chr  "Warner Bros." "Columbia Pictures" "Lucasfilm" "Paramount Pictures" ...
## $ runtime   : num  146 104 124 88 98 95 133 129 127 100 ...
```

```
summary(movies_data[, c("year", "score", "votes", "budget", "gross", "runtime")])
```

```
##      year      score      votes      budget
## Min.   :1980  Min.   :1.90  Min.    :      7  Min.    :      3000
## 1st Qu.:1991  1st Qu.:5.80  1st Qu.:   9100  1st Qu.: 10000000
## Median :2000  Median :6.50  Median :  33000  Median : 20500000
## Mean   :2000  Mean   :6.39  Mean   :  88109  Mean   : 35589876
## 3rd Qu.:2010  3rd Qu.:7.10  3rd Qu.:  93000  3rd Qu.: 45000000
## Max.   :2020  Max.   :9.30  Max.   :2400000  Max.   :35600000
##                NAs    :3      NAs    :3          NAs    :2171
##      gross      runtime
## Min.   :3.090e+02  Min.    : 55.0
## 1st Qu.:4.532e+06  1st Qu.: 95.0
## Median :2.021e+07  Median :104.0
## Mean   :7.850e+07  Mean   :107.3
## 3rd Qu.:7.602e+07  3rd Qu.:116.0
## Max.   :2.847e+09  Max.   :366.0
## NAs    :189      NAs    :4
```

Explanation: `str()` shows the data type of each column. `summary()` gives quick statistics (min, max, average) for the key numeric columns: year, score, votes, budget, gross, and runtime.

3. Data Preprocessing and Cleaning

```
colSums(is.na(movies_data[, c("score", "votes", "budget", "gross", "runtime")]))
```

```
##   score  votes  budget  gross runtime
##      3      3   2171   189      4
```

```
movies_clean <- na.omit(movies_data[, c("name", "genre", "year", "score",
                                         "votes", "budget", "gross",
                                         "runtime", "country")])

nrow(movies_clean)
```

```
## [1] 5435
```

```
sum(duplicated(movies_clean))
```

```
## [1] 0
```

Explanation: `colSums(is.na(...))` counts missing values in the key numeric columns (this dataset commonly has gaps in budget and gross). `na.omit()` removes incomplete rows so our financial analysis isn't skewed by missing data. We also check for duplicate rows.

4. Data Analysis and Statistical Exploration

```
median(movies_clean$score)
```

```
## [1] 6.5
```

```
var(movies_clean$score)
```

```
## [1] 0.9277055
```

```
sd(movies_clean$score)
```

```
## [1] 0.9631747
```

```
genre_counts <- table(movies_clean$genre)
genre_counts
```

```
##
##   Action Adventure Animation Biography Comedy Crime Drama Family
##   1416      327      278      312    1496     400     869        4
##   Fantasy   Horror   Mystery   Romance   Sci-Fi Thriller Western
##      42      254      17        5        6        7        2
```

```
gross_by_genre <- aggregate(gross ~ genre, data = movies_clean, sum)
gross_by_genre <- gross_by_genre[order(-gross_by_genre$gross), ]
head(gross_by_genre, 10)
```

```
##      genre      gross
## 1   Action 237752949398
## 5   Comedy 88514817620
## 3 Animation 77874645616
## 7    Drama 52341119281
## 2  Adventure 43578711908
## 6    Crime 20032912756
## 4 Biography 19096617444
## 10   Horror 14262010058
## 8   Family 2074332587
## 11  Mystery 2004091467
```

```
cor(movies_clean$budget, movies_clean$gross)
```

```
## [1] 0.74041
```

```
cor(movies_clean$score, movies_clean$votes)
```

```
## [1] 0.4737885
```

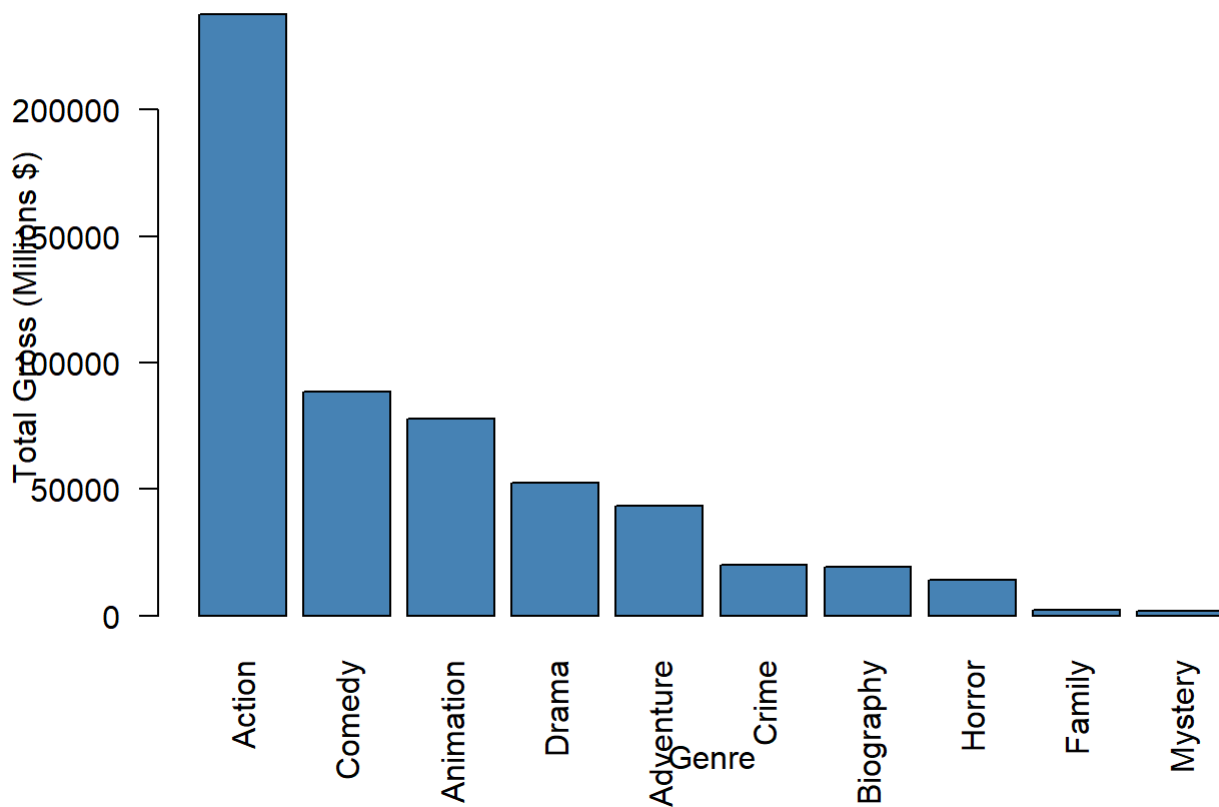
Explanation: `median()`, `var()`, and `sd()` describe how spread out audience scores are across movies. `table()` counts how many movies fall into each genre. `aggregate()` sums total box office gross per genre to find the most profitable ones. `cor()` checks whether bigger budgets lead to bigger box office returns, and whether higher-rated movies attract more votes.

5. Graphs and Plots

```
top_genres <- head(gross_by_genre, 10)

barplot(
  top_genres$gross / 1e6,
  names.arg = top_genres$genre,
  main = "Top 10 Genres by Total Box Office Gross",
  xlab = "Genre",
  ylab = "Total Gross (Millions $)",
  las = 2,
  col = "steelblue"
)
```

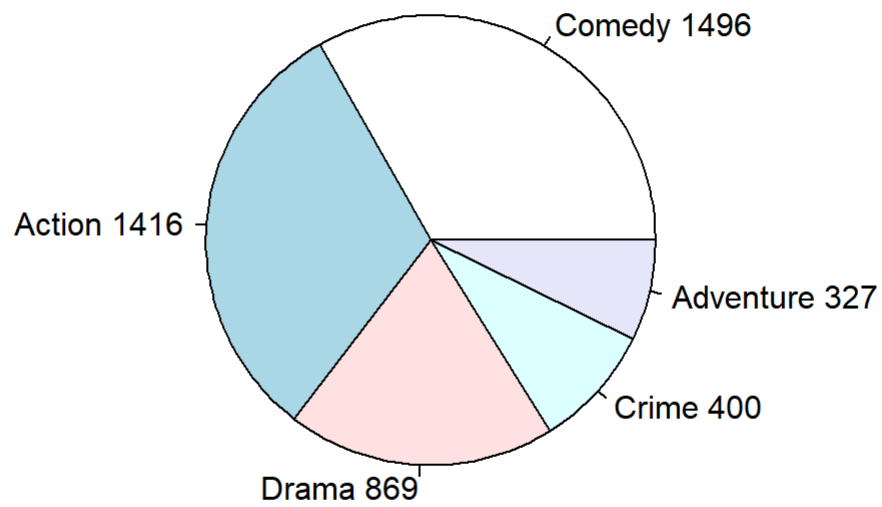
Top 10 Genres by Total Box Office Gross



```
top5_genre_counts <- sort(genre_counts, decreasing = TRUE)[1:5]
```

```
pie(  
  top5_genre_counts,  
  main = "Top 5 Genres by Number of Movies",  
  labels = paste(names(top5_genre_counts), top5_genre_counts)  
)
```

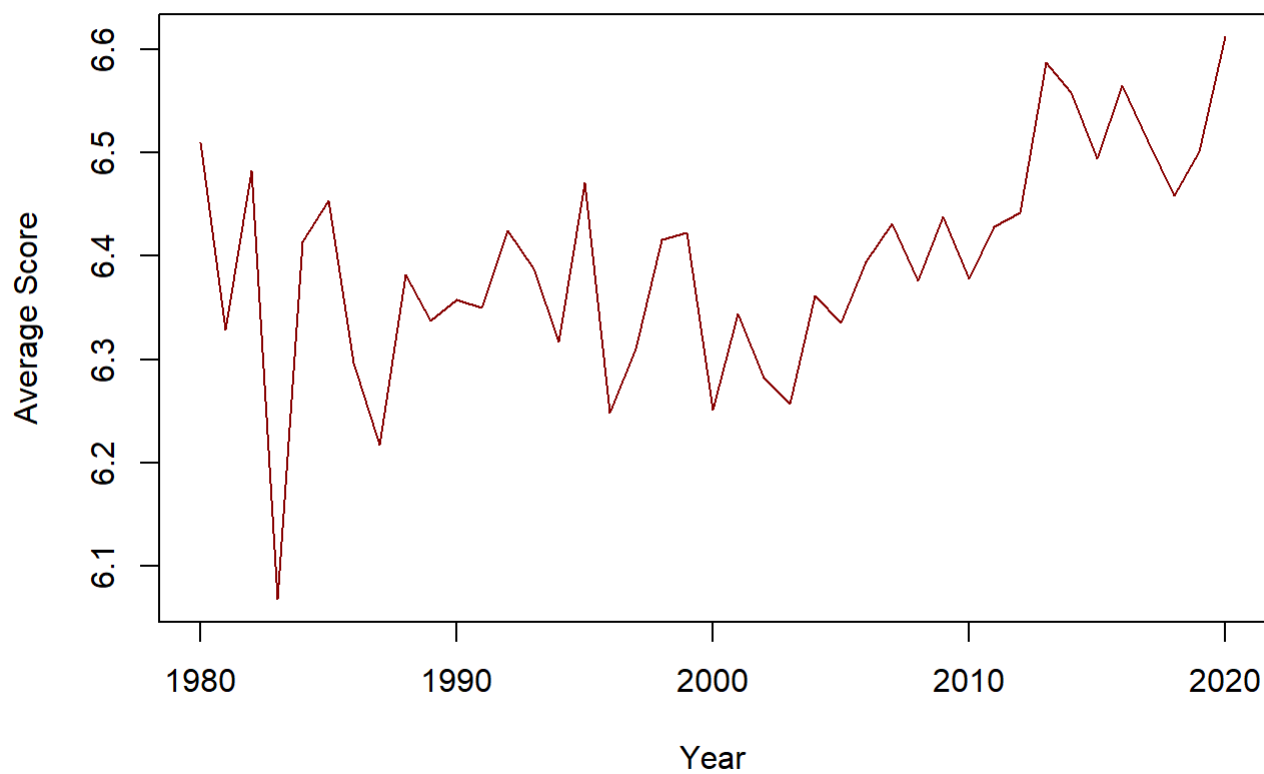
Top 5 Genres by Number of Movies



```
avg_score_by_year <- aggregate(score ~ year, data = movies_clean, mean)

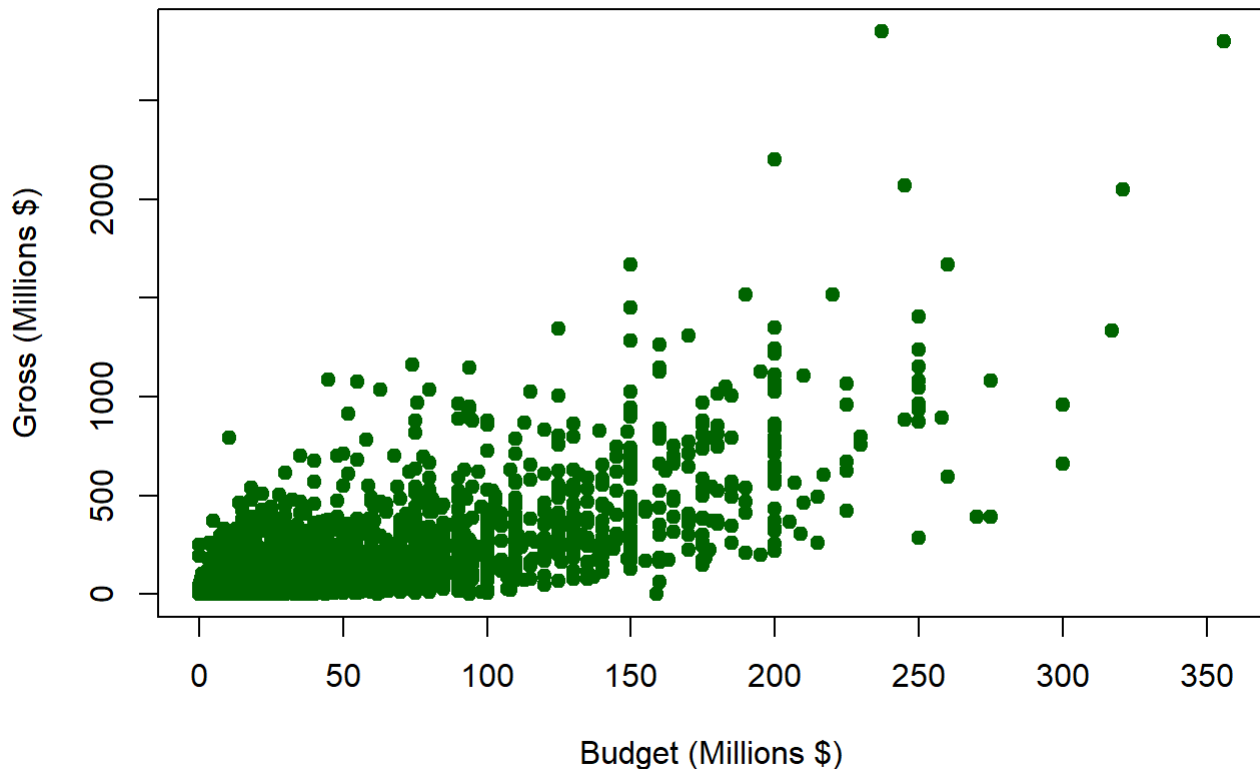
plot(
  avg_score_by_year$year,
  avg_score_by_year$score,
  type = "l",
  main = "Average Movie Score Over Time",
  xlab = "Year",
  ylab = "Average Score",
  col = "darkred"
)
```

Average Movie Score Over Time



```
plot(  
  movies_clean$budget / 1e6,  
  movies_clean$gross / 1e6,  
  main = "Budget vs Box Office Gross",  
  xlab = "Budget (Millions $)",  
  ylab = "Gross (Millions $)",  
  pch = 19,  
  col = "darkgreen"  
)
```

Budget vs Box Office Gross



Explanation: The bar chart shows which genres earned the most overall. The pie chart shows which genres are most common in the dataset. The line chart shows how average audience scores have shifted across years. The scatter plot shows whether higher-budget movies tend to earn more at the box office.

6. Findings and Conclusion

(Fill this in after running the chunks above and looking at your actual numbers/plots — a few example points to build on:)

- The **[Genre Name]** genre generated the highest total box office gross, making it the most commercially successful category.
- The **[Genre Name]** genre had the most movies overall, showing it is the most commonly produced type of film.
- The correlation between budget and gross was **[value]**, indicating a [strong/moderate/weak] relationship — meaning bigger budgets [do/do not] reliably lead to bigger box office returns.
- The correlation between score and votes was **[value]**, suggesting that higher-rated movies [do/do not] tend to attract more viewer engagement.
- Average scores over the years show [an upward/downward/stable] trend, suggesting [audience standards or production quality changed/remained consistent] over time.

Conclusion: This analysis of movie data highlights which genres dominate in popularity versus profitability, and shows that a bigger budget does not always guarantee greater box office success. These insights reflect broader patterns in audience preference and studio investment strategy over time.